

Circular Equal-Mass Hard Rods after Rotation Reduction: Exact Free-Flow Conjugacy, All Events, and Return Times

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Abstract

For N indistinguishable equal-mass rods of length a on a circle of circumference $\ell > Na$, we identify the exact dynamical object on which the usual excluded-volume compression is globally valid. After quotienting constant spatial rotations and gluing elastic collision faces, compression to the free length $L = \ell - Na$ conjugates the hard-rod flow to free phase points modulo common translations and pair permutations.

1 The corrected phase space

The frozen release obstruction identifier is **HEN-0280**. Write $\mathbb{T}_r = \mathbb{R}/r\mathbb{Z}$ and orient it clockwise. A rod is represented by its left endpoint. Choose a cyclic lift $x_1, \dots, x_N \in \mathbb{R}$ satisfying

$$x_{i+1} - x_i \geq a \quad (1 \leq i < N), \quad x_1 + \ell - x_N \geq a. \quad (1)$$

The free gaps are the left sides of (1) minus a ; they are nonnegative and sum to

$$L = \ell - Na > 0.$$

We quotient global constant rotations $x_i \mapsto x_i + r$ and cyclic changes of the first rod. On a contact face, incoming and outgoing velocity labels are glued by the equal-mass elastic reflection. At an intersection of faces, the gluing is generated by the adjacent transpositions in the maximal contact block. The resulting rotation-reduced, unlabeled, collision-glued phase space is denoted by $\mathcal{R}_{N,\ell,a}$.

The free target is

$$\mathcal{F}_{N,L} = ((\mathbb{T}_L \times \mathbb{R})^N) / (S_N \times \mathbb{T}_L), \quad (2)$$

where S_N permutes position–velocity pairs and $c \in \mathbb{T}_L$ sends every y_i to $y_i + c$ but leaves every velocity unchanged. Thus common drift is retained as phase data even though its spatial motion disappears in shape space. No time-dependent Galilean quotient is taken.

2 Compression and complete-flow conjugacy

For a representative satisfying (1), define

$$y_i = x_i - (i-1)a, \quad 1 \leq i \leq N, \quad (3)$$

and reduce the y_i modulo L in (2).

Lemma 1 (Cyclic-start identity). *If the next rod is chosen as the first rod, then the compressed lifted list is*

$$(y_2 + a, \dots, y_N + a, y_1 + L + a).$$

Consequently its class in (2) is unchanged.

Proof. The new lifted endpoint list is $(x_2, \dots, x_N, x_1 + \ell)$. Subtracting $(i-1)a$ gives the displayed list, because $\ell - (N-1)a = L + a$. Modulo L this is a cyclic permutation followed by one common translation by a . A global rod rotation similarly becomes a common translation. Hence (3) is well defined precisely on the reduced source. $\square$

Theorem 1 (Exact shape conjugacy). *Compression induces a bijection*

$$C : \mathcal{R}_{N,\ell,a} \longrightarrow \mathcal{F}_{N,L}.$$

For every $t \in \mathbb{R}$, it conjugates the complete hard-rod flow R_t to the free flow

$$F_t[(y_i, v_i)_{i=1}^N] = [(y_i + tv_i, v_i)_{i=1}^N], \quad CR_t = F_t C. \quad (4)$$

Proof. In cyclic order, (3) converts the free rod gaps exactly into the cyclic gaps between the y_i ; both are nonnegative and sum to L . Conversely, cyclically order any representative of the free points, choose lifts $y_1 \leq \dots \leq y_N \leq y_1 + L$, and put $x_i = y_i + (i-1)a$. Then (1) holds. Common translation changes only the removed rod rotation, a different first point gives the cyclic relation in Lemma 1, and a permutation changes no unlabeled state. Ties are exactly contact faces, where the collision gluing removes the remaining velocity-order ambiguity. This proves bijectivity.

Between contacts, subtracting constants in (3) does not change velocities, so both sides follow (4). At an ordinary equal-mass collision, the rods exchange velocities; letting the two free phase points cross produces the same unlabeled pair. Adjacent transpositions generate every permutation within a simultaneous contact block. Therefore the identity persists through every collision, and the free group defines both forward and backward hard-rod evolution without an event-order convention. $\square$

Jepsen’s equal-mass hard-rod dynamics and free-crossing mechanism are classical [1], as is the excluded-volume setting of the Tonks gas [2]. Our result is a normalized source-model closure and not a claim of literature priority.

References

- [1] D. W. Jepsen, “Dynamics of a Simple Many-Body System of Hard Rods,” *Journal of Mathematical Physics* 6(3) (1965), 405–413, [doi:10.1063/1.1704288](https://doi.org/10.1063/1.1704288).
- [2] L. Tonks, “The Complete Equation of State of One, Two and Three-Dimensional Gases of Hard Elastic Spheres,” *Physical Review* 50 (1936), 955–963, [doi:10.1103/PhysRev.50.955](https://doi.org/10.1103/PhysRev.50.955).